



Genome-wide identification, characterization, and expression analysis of the expansin gene family in Chinese jujube (*Ziziphus jujuba* Mill.)

Lu Hou¹ · Zhiyong Zhang² · Suhan Dou³ · Yadong Zhang³ · Xiaoming Pang¹ · Yingyue Li¹ 

Received: 29 August 2018 / Accepted: 23 September 2018 / Published online: 8 November 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract

Main conclusion 30 expansin genes were identified in the jujube genome. Phylogenetic analysis classified expansins into 17 subgroups. Closely related expansins share a conserved gene structure. *ZjEXPs* had different expression patterns in different tissues.

Plant-specific expansins were first discovered as pH-dependent cell-wall-loosening proteins involved in diverse physiological processes. No comprehensive analysis of the expansin gene family has yet been carried out at the whole genome level in Chinese jujube (*Ziziphus jujuba* Mill.). In this study, 30 expansin genes were identified in the jujube genome. These genes, which were distributed with varying densities across 10 of the 12 jujube chromosomes, could be divided into four subfamilies: 19 *ZjEXPAs*, 3 *ZjEXPBs*, 1 *ZjEXLA*, and 7 *ZjEXLBs*. Phylogenetic analysis of expansin genes in *Arabidopsis*, rice, apple, grape, and jujube classified these genes into 17 subgroups. Members of the same subfamily and subgroup shared conserved gene structure and motif compositions. Homology analysis identified 20 homologous gene pairs between jujube and *Arabidopsis*. Further analysis of *ZjEXP* gene promoter regions uncovered various growth, development and stress-responsive *cis*-acting elements. Expression analysis and transcript profiling revealed that *ZjEXPs* had different expression patterns in different tissues at various developmental stages. *ZjEXPA4* and *ZjEXPA6* were highly expressed in young fruits, *ZjEXPA3* and *ZjEXPA5* were significantly expressed in flowers, and *ZjEXPA7* was specifically expressed in young leaves. The results of this study, the first systematic analysis of the jujube expansin gene family, can serve as a strong foundation for further elucidation of the physiological functions and biological roles of jujube expansin genes.

Keywords Expansin · Gene family · Genome-wide analysis · Expression profile · Chinese jujube

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00425-018-3020-9>) contains supplementary material, which is available to authorized users.

✉ Yingyue Li
yingyueli@bjfu.edu.cn

¹ National Engineering Laboratory for Tree Breeding, College of Biological Sciences and Technology, Beijing Forestry University, Beijing 100083, China

² Beijing Key Laboratory of Ornamental Plants Germplasm Innovation and Molecular Breeding, National Engineering Research Center for Floriculture, Beijing Laboratory of Urban and Rural Ecological Environment, Key Laboratory of Genetics and Breeding in Forest Trees and Ornamental Plants of Ministry of Education, School of Landscape Architecture, Beijing Forestry University, Beijing 100083, China

³ Henan Longyuan Flowers & Trees Co., Ltd., Xuchang 461000, China

Abbreviations

EXLA(B) Expansin-like A(B)

EXPA(B) α -Expansin (β -Expansin)

Introduction

Expansins are a group of proteins that regulate cell-wall extension by disrupting non-covalent bonding between the wall matrix polysaccharides and cellulose microfibrils through a non-enzymatic pH-dependent mechanism (Cosgrove 2005, 2015). Expansins are widespread in plants, and related proteins have also been discovered in non-plant organisms, such as fungi, bacteria, mollusks, and nematodes, that probably contribute to digest plant biomass (Georgelis et al. 2015; Tovar-Herrera et al. 2015). Expansins are often expressed in tissue-specific patterns and play significant roles in diverse biological processes associated with cell-wall modification, including root development (Che et al.

2016), leaf growth (Kuluev et al. 2017), flower development and abscission (Saito et al. 2015; Tsuchiya et al. 2015), fruit ripening and dehiscence (Palapol et al. 2015), fruit firmness and softening (Perini et al. 2017), seed yield and germination (Yan et al. 2014; Chen et al. 2016), and biotic and abiotic stress responses (Perini et al. 2017).

Typical expansins in plants are 250–275 amino acids long and comprise two domains (1 and 2) preceded by a signal peptide of approximately 20–30 amino acid residues (Yennawar et al. 2006). The N-terminal domain 1 is a double-psi beta-barrel (DPBB), which is homologous to glycoside hydrolase family 45 (GH45) and includes a conserved HFD motif and a number of conserved cysteines. The C-terminal domain 2 is distantly related to grass group-2 pollen allergens (G2As) (Cosgrove 2000; Yennawar et al. 2006). Expansins, which are encoded by a multi-gene family, are divided into four families designated as α -expansin (EXPA), β -expansin (EXPB), expansin-like A (EXLA), and expansin-like B (EXLB) on the basis of phylogenetic analyses and standardized nomenclature (Kende et al. 2004). EXPA and EXPB families are the two major subfamilies and have been experimentally demonstrated to exert expansin wall-loosening activities, whereas EXLA and EXLB families are identified only by their conserved amino acid sequences, with far less known about their precise functions (Cosgrove et al. 1997; Sampedro and Cosgrove 2005). *AtEXLA2* in *Arabidopsis* and *AraEXLB8* in *Arachis* has recently been experimentally demonstrated to be involved in root and hypocotyl lengths and stress response, respectively (Boron et al. 2015; Guimaraes et al. 2017).

Chinese jujube (*Ziziphus jujuba* Mill.) ($2n=2x=24$), a member of the Rhamnaceae family and native to China, is one of the most economically important fruit trees in Asia (Liu 2006; Liu and Wang 2009). Jujube fruits are very rich in calories, nutritive substances and biologically active components, including vitamin C, minerals (calcium and iron), carbohydrates, and cyclic adenosine monophosphate; they are thus used extensively in traditional Chinese medicine (Lu et al. 2010; Gao et al. 2013). Because it is widely adapted, early bearing, long lived and easy to manage with multiple uses that mesh well with long-term intercropping systems, jujube is considered an ideal economic crop for arid and semiarid temperate zone areas, where common fruit trees grow poorly (Liu 2003, 2006). Draft genome sequences of the fresh jujube cultivar ‘Dongzao’ (437.7 Mb) and the dry jujube cultivar ‘Junzao’ (351.1 Mb) have been recently reported (Liu et al. 2014; Huang et al. 2016). The availability of these sequences, which offer rich genomic resources for deep molecular genetic research, provides an excellent opportunity to investigate all genes belonging to specific gene families in jujube.

Although expansins have been analyzed in organs of many plant species, such as *Arabidopsis* (Lee et al. 2001),

rice (Lee et al. 2001), maize (Zhang et al. 2014b), wheat (Li et al. 2016), soybean (Zhu et al. 2014), tobacco (Ding et al. 2016), tomato (Lu et al. 2016), apple (Zhang et al. 2014a), and grape (Dal Santo et al. 2013), the features and functions of expansin gene family in jujube are currently unavailable. In this study, we identified 30 expansin genes representing all four families. The phylogenetic relationships, gene structures, and homologous gene pairs of all identified expansin genes, as well as the *cis*-acting elements in their promoters were then systematically analyzed. We also investigated the transcription and expression profiles of jujube expansin genes in different tissues. Our results provide a foundation for future research on the evolutionary of expansin genes in plants. Our findings should advance understanding of the biological functions and roles of expansin genes in jujube growth and development, especially fruit ripening, and help to the further genetic modification of jujube to improve fruit tree quality.

Materials and methods

Plant materials

The *Z. jujuba* Mill. ‘Dongzao’ from the National Foundation for Improved Cultivar of Chinese Jujube (Cangzhou, China) was used for organ-specific expression analysis. Young leaves, mature leaves, shoots, flowers, young fruits, white-mature fruits, half-red fruits, and full-red fruits of ‘Dongzao’ were collected separately from three jujube trees. The fruits were collected at four different development stages from July 5 to September 22. All collected tissues were immediately frozen in liquid nitrogen and stored at -80°C until needed for RNA isolation.

Identification of the expansin genes in jujube

Expansin protein sequences of *Arabidopsis* were downloaded from the databases of the *Arabidopsis* Information Resource (<http://www.arabidopsis.org/>). First, all sequences were used as query sequences to perform tblastn with 1×10^{-5} cut-off *E* value against the Dongzao reference genome sequence (version 1.1, Liu et al. 2014). The highly matched sequences were reorganized and merged. Then, all protein sequences of putative expansin genes were scanned for the expansin conserved domain using InterProScan (<http://www.ebi.ac.uk/interpro/>, Finn et al. 2017). The sequences lacking expansin domains were rejected in this analysis. Finally, we further verified these sequences using EST database in NCBI.

Protein properties, chromosomal location, and gene structure analysis

The ExPaSy (<http://expasy.org/>, Artimo et al. 2012) was used to compute the molecular weight (MW) and isoelectric point (pI) for each protein sequence. Each expansin gene was mapped to the chromosomes by the MapInspect software according to their coordinates on the genome. The coding domain sequences (CDS) and DNA sequences of the expansin genes were used to predict the exon–intron structures using the online tool GSDS (version 2.0: <http://gsds.cbi.pku.edu.cn>, Hu et al. 2015). The conserved motifs were identified using the online program MEME (version 4.10.2: <http://meme.sdsc.edu/meme/intro.html>, Bailey et al. 2009), with optimum motif width ≥ 6 bp and ≤ 200 bp and maximum number of motifs 15.

Sequence alignment, phylogenetic tree construction, and homology analysis

Multiple sequence alignment of the amino acid sequences was conducted by the DNAMAN software (version 6.0). The full-length amino acid sequences were aligned with the ClustalW program (Thompson et al. 1994) and a phylogenetic tree was constructed by the MEGA software (version 6, <http://www.megasoftware.net/>, Tamura et al. 2013) using the neighbour-joining (NJ) method with 1000 bootstrap replicates. The expansin genes in jujube and *Arabidopsis* were clustered using the entire protein sequences by the OrthoMCL software (Li et al. 2003) and the homologous relationships were drawn using the Circos software (Krzywinski et al. 2009).

Analysis of *cis*-acting elements in *ZjEXP* gene promoter regions

The promoter regions (1500 bp sequence upstream of the genomic DNA sequence) of the *ZjEXPs* genes were identified by searching the Dongzao genome database (<http://www.genoscope.cns.fr/brassicanaapus/>), and they were then analyzed using PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) to predict the *cis*-acting elements (Lescot et al. 2002).

Transcriptome analysis of the jujube expansin genes

The transcriptome data of gene expression in six different tissues including root, leaves, flower, stem, branch, and fruit of Dongzao were downloaded from NCBI sequence read archive (SRA) with accession code SRP046073 (Liu et al. 2014). Raw sequences were filtered as described by Song et al. (2017). SRA files were converted to FASTQ files using the SRA toolkit (version 2.8.2). After the file conversion,

FastQC program was employed to check read quality and Trimmomatic (version 0.32) was used to remove low-quality portions of reads with parameter settings below: SLIDING WINDOW: WINDOW SIZE = 4 bp, QUALITY = 15; LEADING = 3; TRAILING = 3; MINLEN = 15. Trimmed paired reads were aligned to a Bowtie2-indexed jujube reference genome (version 1.1) with TopHat (version 2.1.1, Trapnell et al. 2012). The bam (aligned read) files were processed to analyze gene expression quantitatively using Cufflinks (version 2.2.1) with default parameters. The levels of gene expression were estimated by FPKM values (fragments per kilobase per million mapped reads, Li and Dewey 2011). The raw FPKM values were first normalized by logarithmic method and a heat map was constructed to show the different expression profiles by the Heml software (version 1.0).

RNA isolation and qRT-PCR analysis

The total RNA of different tissues was isolated by the E.Z.N.A.TM Plant RNA Kit (Omega Bio-Tec, Norcross, GA, USA). The RNA was analyzed by agarose gel electrophoresis, and the concentration and purity were checked by NanoDrop 2000 (Thermo Fisher Scientific, Waltham, MA, USA). The first cDNA strand was synthesized from 1 μ g total RNA using the FastQuant RT Kit (with gDNase) (Tiangen, Beijing, China) according to the manufacturer's instructions. The cDNA were then diluted ten times and stored at -20°C for the subsequent qRT-PCR. The specific primers were designed with Beacon Designer 7.9 for qRT-PCR analysis. *ZjACT* reference gene was used as an internal control (Bu et al. 2016). Primer details primers are listed in Table S1. qRT-PCR was conducted using SYBR green (Aidlab Biotechnology, Beijing, China) on the Analytik Jena qTower2.2 instrument. The 20 μ L reaction contained 1 μ L of cDNA, 400 nM of each primer, and 10 μ L of SYBR Green mix, according to the manufacturer's instructions. The PCR cycle was as follows: 94°C for 3 min, 40 cycles of 94°C for 20 s, 55°C for 20 s, and 72°C for 20 s. The fluorescence was measured during the extension step. The specificity of the amplicon for each primer pair was verified by melting curve analysis. Three biological replicates were utilized for each experiment, and each replicate was measured in triplicate. The relative gene expression levels were calculated according to the $2^{-\Delta\Delta\text{CT}}$ method (Livak and Schmittgen 2001).

Results

Identification of the expansin gene family in jujube

To identify expansin genes in the jujube genome, BLAST searches were performed using the corresponding sequences from *Arabidopsis* as queries. After filtering on the basis of

E value ($< 1 \times 10^{-5}$) and bias criteria, 43 sequences were identified as putative expansin genes in jujube. Scanning of the expansin conserved domain uncovered 30 expansin genes with two conserved domains in their protein sequences in the jujube genome. In accordance with the nomenclature methods for members of expansin superfamily proposed by Kende et al. (2004), all jujube expansin genes were sequentially named on the basis of family identity and chromosomal position (Table 1). The identified *ZjEXP* genes encoded proteins ranging from 193 (*ZjEXPA10*) to 295 (*ZjEXPA19*) amino acids (aa) in length and had predicted molecular weights (MWs) of 21.23–32.55 kDa. All members of *ZjEXPA* (except *ZjEXPA13*), *ZjEXPB*, and *ZjEXLA* subfamilies had pI values above 7.0, whereas pI values of most members of *ZjEXLB* (except *ZjEXLB2* and *ZjEXLB4*) were below 7.0.

The chromosomal distribution map of jujube expansin genes was generated on the basis of jujube genome data

(Fig. 1). Of the 30 expansin genes in jujube, 29 were successfully mapped to 10 of the 12 jujube chromosomes, while one expansin gene, *ZjEXPA19*, was only located on an unplaced genomic scaffold. Chr11 contained the largest number (11) of expansin genes and harbored all the EXLB genes. Only one expansin gene each was found on chr5 and chr7; two each were found on chr1, chr2, chr3, chr8, and chr10, and three each were assigned to chr4 and chr9.

Sequence alignment and phylogenetic analysis

To obtain detailed information on each jujube expansin gene and superfamily, 30 jujube expansin genes were subjected to multiple sequence alignment (Fig. 2). Most of the expansins consisted of three parts, namely, a signal peptide and domains 1 and 2. The domain 1 was more conserved in amino acid sequence than domain 2. All EXPBs and the majority of EXPA members (excluding *ZjEXPA8*,

Table 1 Description of jujube expansin family genes

Gene name	Gene ID	RefSeq ID	Location	Strand	ORF (aa)	CDS (bp)	MW (KDa)	pI
<i>ZjEXPA1</i>	LOC107422842	XM_016032355.1	chr1:17202460..17206669	Minus	257	774	27.78	9.39
<i>ZjEXPA2</i>	LOC107425594	XM_016035605.1	chr1:23386740..23388645	Plus	252	759	27.74	7.51
<i>ZjEXPA3</i>	LOC107411032	XM_016018540.1	chr2:8826319..8829123	Plus	258	777	27.94	9.37
<i>ZjEXPA4</i>	LOC107411451	NM_001323858.1	chr2:16642656..16644137	Plus	254	765	27.21	8.61
<i>ZjEXPA5</i>	LOC107413433	XM_016021381.1	chr3:8497464..8498850	Minus	285	858	30.94	9.46
<i>ZjEXPA6</i>	LOC107414006	NM_001323865.1	chr3:18313625..18315175	Minus	253	762	26.91	8.32
<i>ZjEXPA7</i>	LOC107415875	XM_016024281.1	chr4:8229498..8230630	Plus	261	786	28.63	8.9
<i>ZjEXPA8</i>	LOC107416179	XM_016024637.1	chr4:11271529..11274159	Minus	268	807	29.15	9.02
<i>ZjEXPA9</i>	LOC107419239	XM_016027983.1	chr5:25014160..25017103	Minus	251	756	26.89	8.88
<i>ZjEXPA10</i>	LOC107423099	XM_016032616.1	chr7:20833381..20835672	Plus	193	582	21.23	8.88
<i>ZjEXPA11</i>	LOC107427332	XM_016037713.1	chr9:18110002..18111266	Plus	237	714	25.69	8.61
<i>ZjEXPA12</i>	LOC107427516	XM_016037895.1	chr9:19921924..19925261	Minus	238	717	25.61	8.92
<i>ZjEXPA13</i>	LOC107428867	XM_016039460.1	chr10:9787299..9788782	Plus	253	762	28.14	5.78
<i>ZjEXPA14</i>	LOC107428959	NM_001323883.1	chr10:10891272..10893298	Minus	250	753	26.63	9.33
<i>ZjEXPA15</i>	LOC107430151	XM_016040951.1	chr11:3649968..3651634	Minus	257	774	28.52	9.27
<i>ZjEXPA16</i>	LOC107430152	XM_016040952.1	chr11:3663216..3664880	Minus	257	774	28.52	9.27
<i>ZjEXPA17</i>	LOC107430635	XM_016041499.1	chr11:11558364..11561407	Minus	258	777	27.94	9.57
<i>ZjEXPA18</i>	LOC107431073	XM_016041948.1	chr11:15188139..15189463	Minus	255	768	27.81	8.64
<i>ZjEXPA19</i>	LOC107403701	XM_016010615.1	chrUn(scaffold1071):26184..30325	Minus	295	888	32.55	9.77
<i>ZjEXPB1</i>	LOC107417370	XM_016025988.1	chr4:29688336..29689500	Plus	206	621	22.48	8.38
<i>ZjEXPB2</i>	LOC107424171	XM_016033911.1	chr8:4743612..4761120	Minus	240	723	26.72	9.56
<i>ZjEXPB3</i>	LOC107424174	XM_016033913.1	chr8:4802312..4815135	Minus	250	753	26.69	8.74
<i>ZjEXLA1</i>	LOC107427312	NM_001323882.1	chr9:17754101..17756095	Minus	259	780	27.96	8.7
<i>ZjEXLB1</i>	LOC107430373	XM_016041206.1	chr11:8136780..8138110	Minus	250	753	27.75	6.2
<i>ZjEXLB2</i>	LOC107430374	XM_016041207.1	chr11:8177039..8178353	Minus	250	753	27.91	7.48
<i>ZjEXLB3</i>	LOC107430372	XM_016041205.1	chr11:8219563..8221201	Minus	250	753	27.86	6.8
<i>ZjEXLB4</i>	LOC107430389	XM_016041228.1	chr11:8773788..8775415	Plus	249	750	27.63	7.49
<i>ZjEXLB5</i>	LOC107430395	XM_016041230.1	chr11:9004435..9005588	Plus	196	591	21.57	5.35
<i>ZjEXLB6</i>	LOC107430443	XM_016041275.1	chr11:9568626..9570170	Plus	257	774	27.89	4.88
<i>ZjEXLB7</i>	LOC107430440	XM_016041273.1	chr11:9576255..9577726	Minus	257	774	27.78	6.11

Fig. 1 Chromosomal distribution of jujube expansin genes

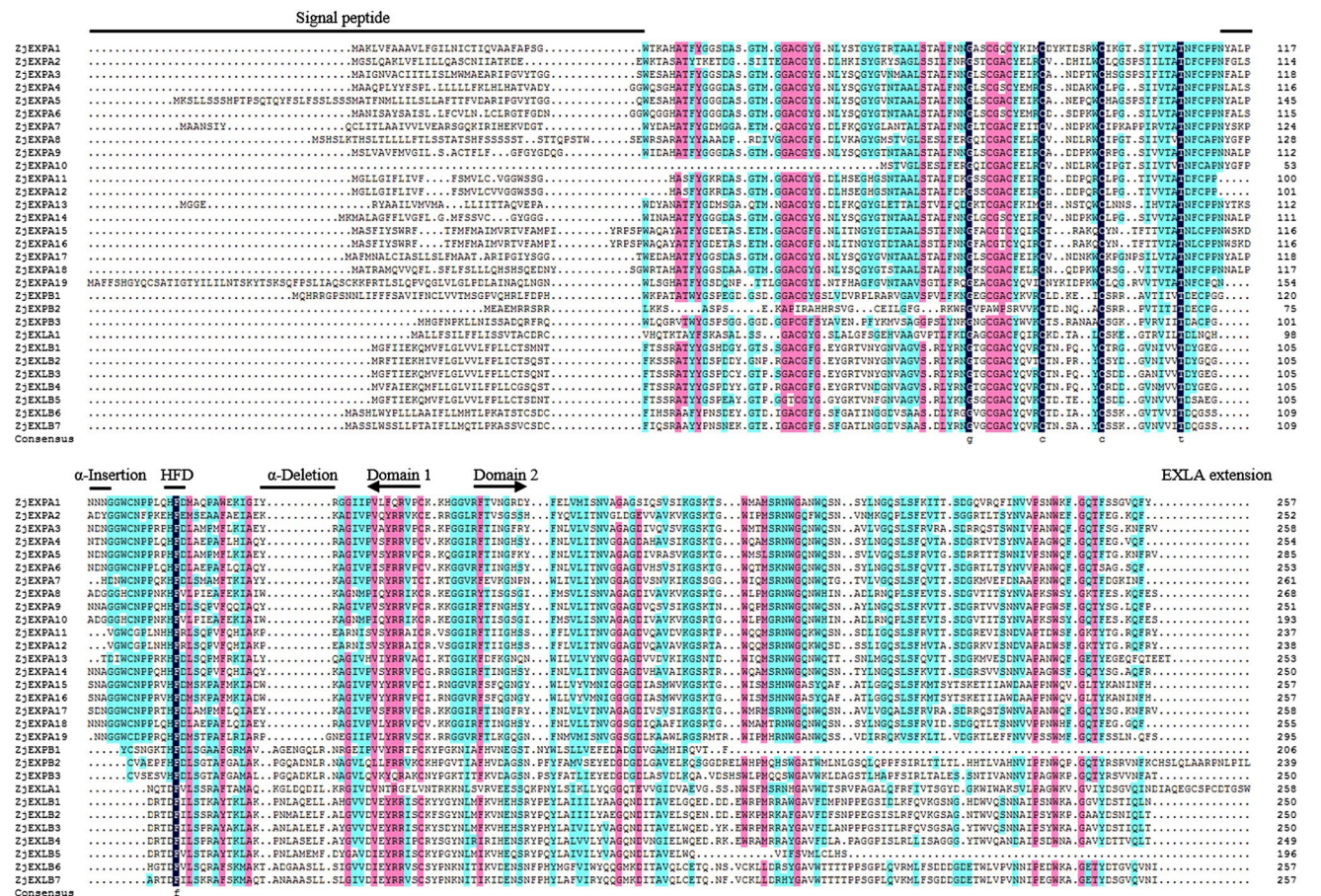
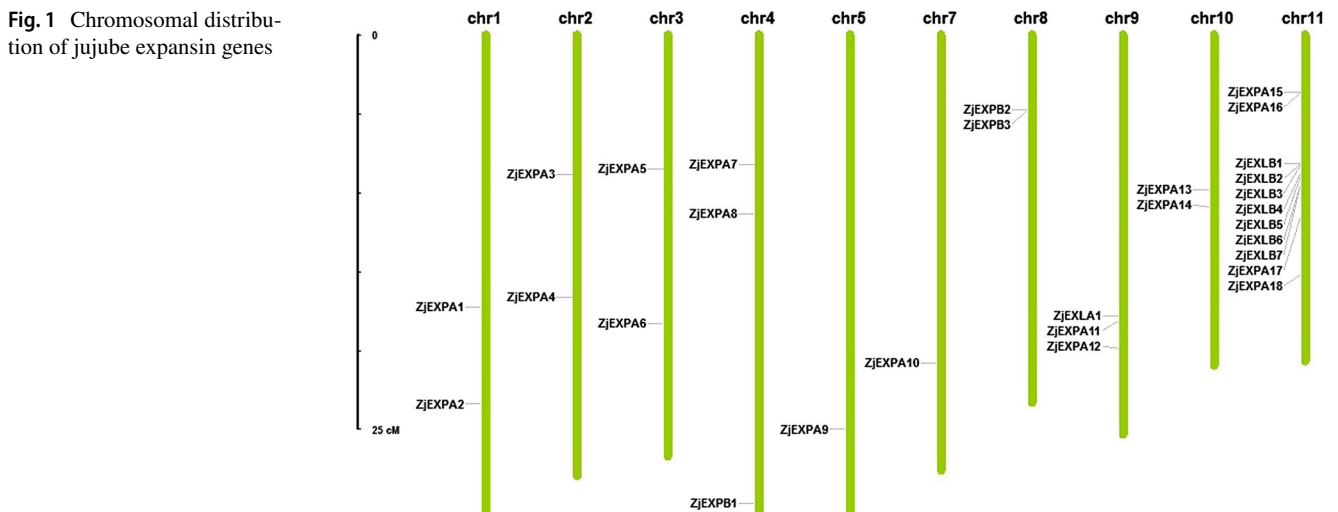


Fig. 2 Multiple sequence alignment of the jujube expansin protein sequences was obtained by the DNAMAN software. The family-specific conserved motifs and active sites are indicated

ZjEXPA10, *ZjEXPA11*, and *ZjEXPA12*) contained a conserved HFD motif in domain 1, which characterize the large insertion (α -insertion) and deletion (α -deletion) unique to *ZjEXPA* subfamily. Of the eight expansin-like protein sequences with an incompletely conserved HFD motif, one was classified into the EXLA subfamily based on the presence of a characteristic EXLA extension at the C-terminus, with the remaining ones classified into EXLB subfamily.

To evaluate evolutionary relationships of expansin proteins, complete protein sequences of 191 expansins from five angiosperm species, namely, a monocot (rice) and four dicots (*Arabidopsis*, apple, grape, and jujube), were subjected to phylogenetic analysis using MEGA (version 6). As shown in the phylogenetic tree in Fig. 3, all expansin genes were divided into four subfamilies: EXPA, EXPB, EXLA, and EXLB, and members of the *ZjEXPA*, *ZjEXPB*, *ZjEXLA*, and *ZjEXLB* subfamilies were clustered together.

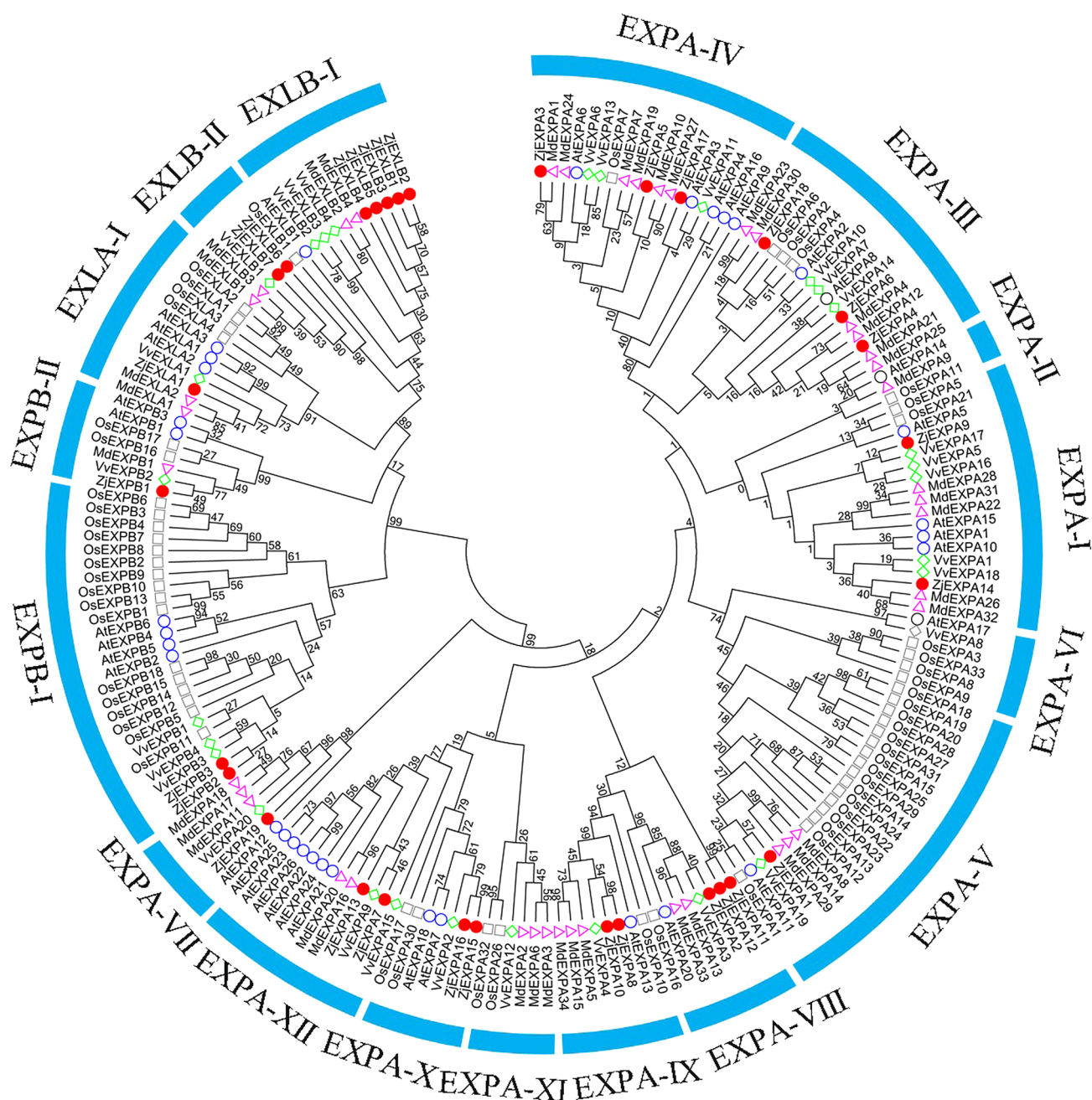


Fig. 3 Phylogenetic analysis of the expansin genes in *Arabidopsis* (*AtEXP*), rice (*OsEXP*), apple (*MdEXP*), grape (*VvEXP*), and jujube (*ZjEXP*). The subgroups are marked by color bars

was the largest subfamily, and EXLA and EXLB were the smallest. The four expansin gene subfamilies were further divided into 17 subgroups according to the grouping rules used for *Arabidopsis*, rice, and pine expansin gene families (Sampedro and Cosgrove 2005). The EXPA-II, EXPA-VI, and EXPA-XI subgroups did not contain any jujube expansin genes. We found that EXPB-I subgroup, containing 25 members, constituted the largest subgroup, while the smallest subgroup, EXPA-II, comprised only three genes.

Gene structure and putative motif identification

When interpreting phylogenetic relationships within gene families, a structural analysis can provide valuable information concerning duplication events. We, therefore, generated exon–intron structures based on the corresponding genome and coding sequences (Fig. 4). The number of exons in the jujube expansin gene family varied from two (*ZjEXPA7*, *ZjEXPA8*, *ZjEXPA10*, and *ZjEXPA13*) to five (*ZjEXLA1*, *ZjEXLB6*, and *ZjEXLB7*), with members within each subgroup or subfamily having similar exon–intron structures. In the EXPA subfamily, 14 members had three exons; four had two exons, and one member (*ZjEXPA19*) had four. The EXPB subfamily had three exons with the exception of *ZjEXPB3*, which contained four. Members of the *ZjEXLA* subfamily possessed five exons, while *ZjEXLB* genes—except for *ZjEXLB5*, which had three—possessed four or five.

We used MEME to detect conserved motifs in the jujube expansin family and found 15 distinct motifs (Fig. 5). Schematic diagrams of these jujube expansin motifs are given in Table S2. Motifs 2 and 10 specified the well-conserved N-terminal domain 1, and motifs 4, 5, and 11 consisted of the well-conserved C-terminal domain 2. At least two of these five motifs were shared among the jujube expansin genes. In addition, motifs 1 and 10 were characteristic of the two conserved domains (Table S2). Genes from the same subfamily, especially within subgroups, were generally characterized by a similar motif type and distribution. Motifs 1, 9, 12, 13, and 15 were only detected in the *ZjEXPA* subfamily, and motif 10 was only identified in the *ZjEXPB* subfamily. The two *ZjEXPB* subgroups EXPB-I and EXPB-II contained motifs 8 and 14, respectively, which differed from each other. Motif 7 was only found in *ZjEXLA* and *ZjEXLB*, whereas motif 5 was only present in *ZjEXLB*. Motif 3 was detected in all *ZjEXPs* except for *ZjEXLA* genes. These results indicate that the *ZjEXPA* subfamily differs significantly from the other three subfamilies in terms of gene structure as well as motif composition.

Homology analysis of jujube and *Arabidopsis* expansin gene families

Because *Arabidopsis* is the most important model plant and many of its expansin genes have been functionally well characterized, we performed a homology analysis of jujube

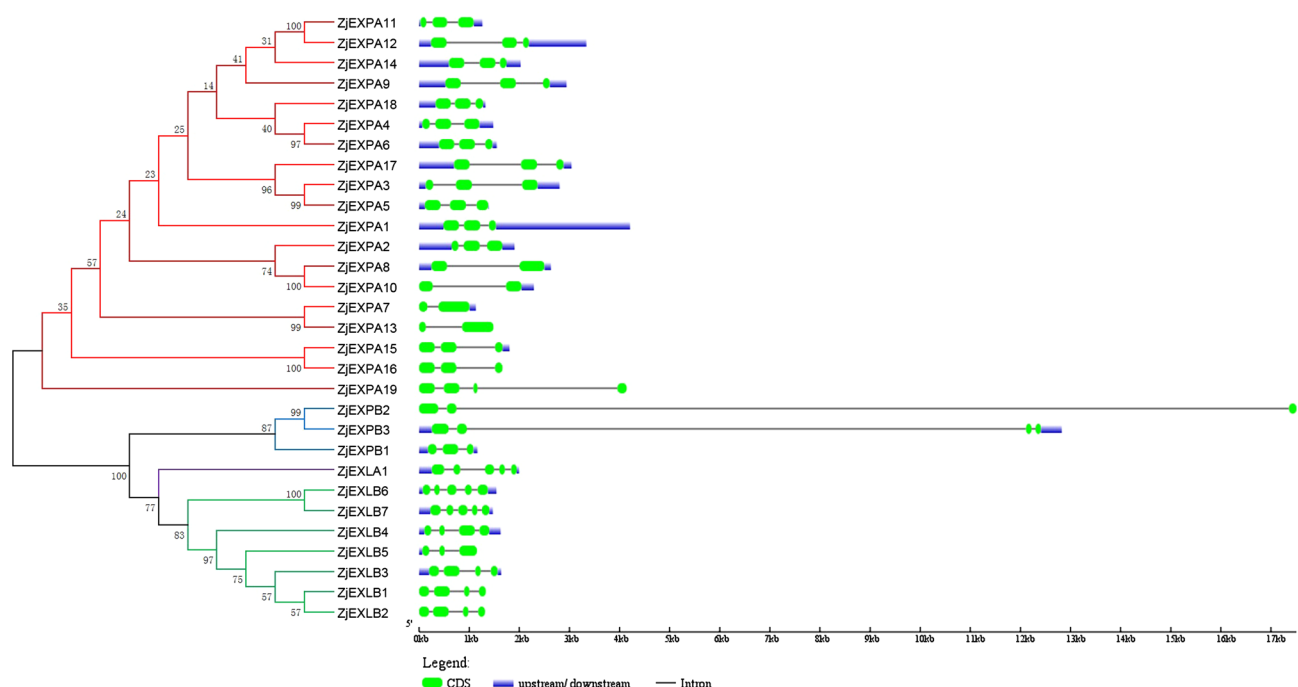


Fig. 4 Phylogenetic relationships and structural analysis of jujube expansin genes. Introns and exons are represented by black lines and green boxes, respectively. Lengths of introns and exons of each expansin gene are displayed proportionally

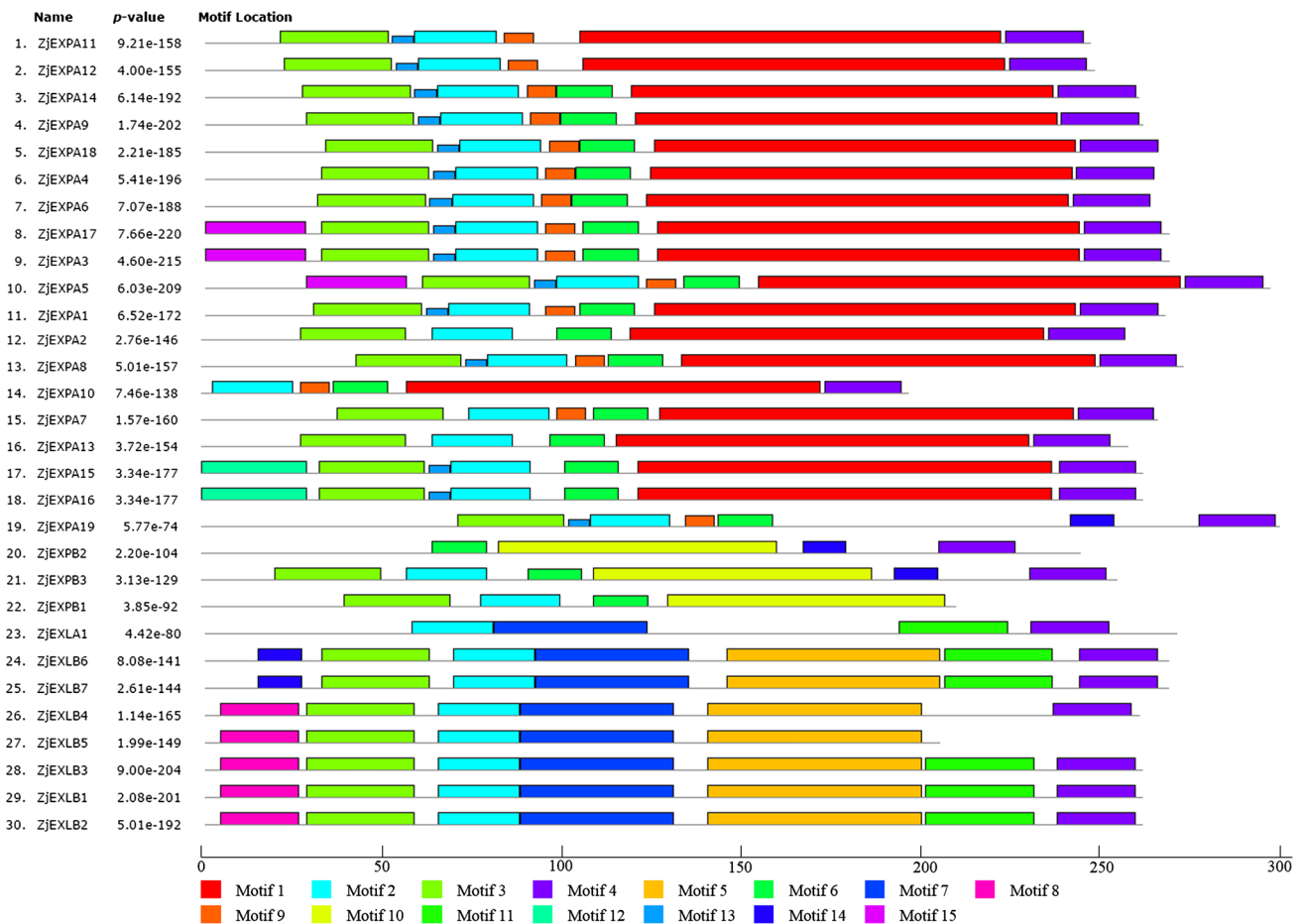


Fig. 5 Putative motifs of each jujube expansin gene by MEME

and *Arabidopsis* expansin genes to predict the function of *ZjEXPs* (Fig. 6). Because the length of its scaffold was too short to include in the homology map, *ZjEXPA19* was excluded from the homology analysis. The analysis revealed the presence of 20 homologous gene pairs between jujube and *Arabidopsis* as well as six gene pairs in jujube and six in *Arabidopsis*.

Detection of *cis*-acting elements in *ZjEXP* gene promoter regions

As detailed in Tables 2 and S3, *cis*-acting elements in the promoter regions of plant expansin genes are extremely diverse. We detected 102 types of *cis*-acting elements, 27 with unknown functions, in the promoters of jujube expansin genes. The most abundant elements in *ZjEXP* gene promoter regions were light-responsive elements, which were represented by up to 34 types (Table S3). Each promoter possessed 5–17 types, which suggests that *ZjEXPs* are differentially regulated in the presence of light (Table 2). Ten types of hormone-responsive elements were detected. The GARE

motif, which is involved in gibberellin (GA) response, was the most abundant hormone-responsive element in the 16 *ZjEXP* gene promoters. In addition, *ZjEXPA19* did not contain any hormone-responsive regulatory elements. An AuxRR core and a TGA box were found in the promoter regions of *ZjEXPA* and *ZjEXPB* genes, which indicate that auxin may regulate these genes. Moreover, nine types of environmental stress-related elements were identified in *ZjEXP* promoters. ARE, Box-W1, HSE, MBS, and TC-rich repeats elements were present in over half of the identified 30 jujube expansin genes. The fourth important type of *cis*-acting elements in the promoter regions of jujube expansin genes was development-related elements. The *skn-1* motif (required for endosperm expression) was the most abundant regulatory element of the nine types and was located in the 24 *ZjEXP* gene promoters. Twenty jujube expansin genes contained the circadian control element (circadian), which suggests that these genes have a distinct expression pattern during diurnal cycle. Notably, the seed-specific regulation-related *cis*-acting element (RY element) was detected in *ZjEXPA7* and *ZjEXPA14* (Table S3).

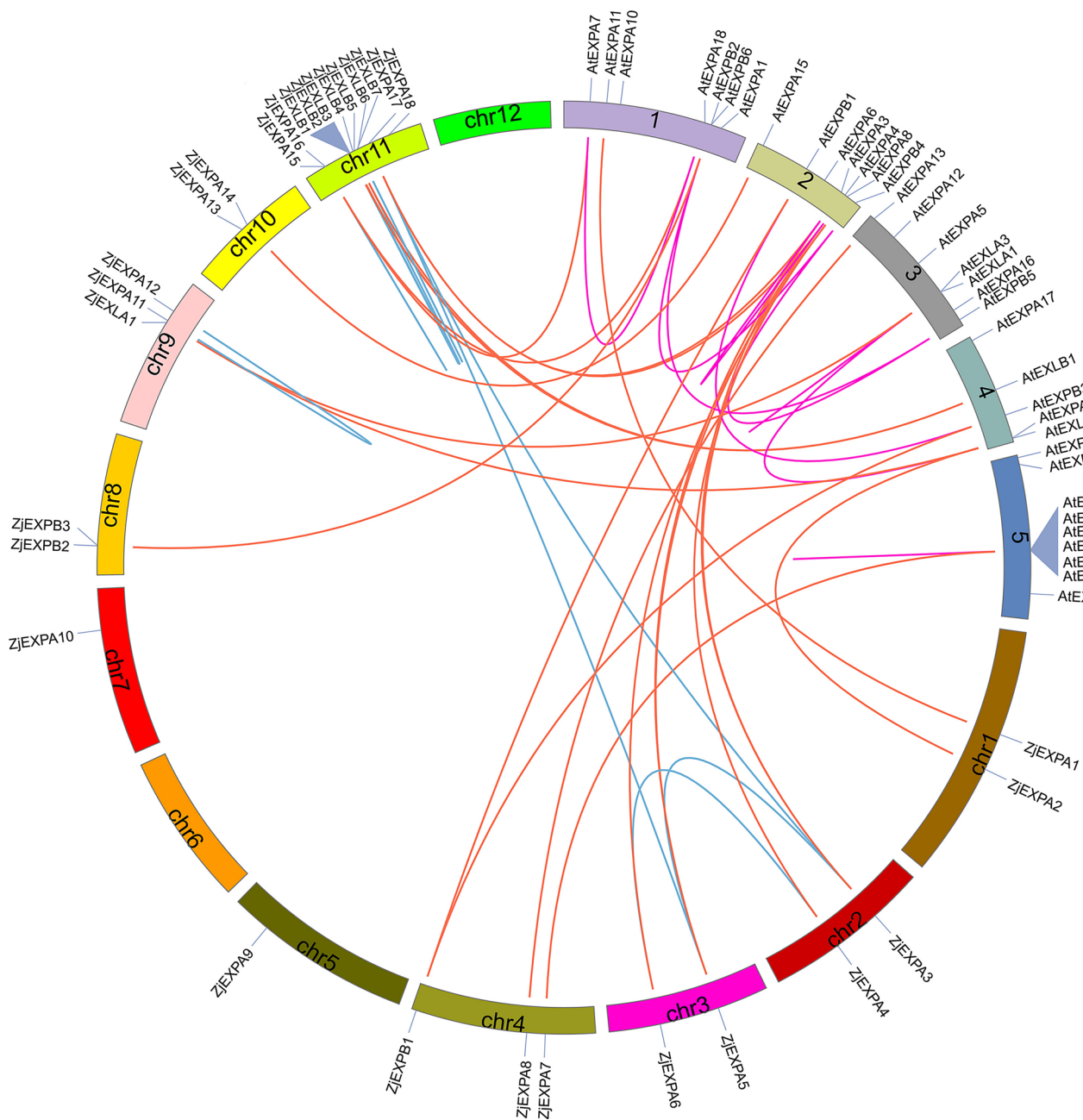


Fig. 6 Homology analysis of expansin genes between jujube and *Arabidopsis*. Red, purple, and blue lines indicate homologous gene pairs between jujube and *Arabidopsis*, and *Arabidopsis* and jujube

Expression patterns of jujube expansin genes revealed by transcriptome analysis

To analyze expression profiles of individual jujube expansin genes, we constructed a hierarchical clustering heat map showing the expression levels of jujube expansin genes in different tissues (Fig. 7). According to the hierarchical clustering (dendrogram), the expression patterns of jujube expansin genes were divided into five groups. All genes from Group I were rarely expressed in fruits, but, except

for *ZjEXPA7*, were highly detected in stems and branches. *ZjEXPA8* and *ZjEXPA10* had similar expression patterns, indicating that these genes have similar biological functions in the same environment. *ZjEXP13* and *ZjEXP16* from Group II were not expressed in any tissues. Group III genes were mainly expressed in leaves, flowers, and branches. *ZjEXLB3* and *ZjEXLB7* exhibited similar expression patterns, suggesting that they may play similar roles in the same biological processes. The expression levels of Group IV and V genes were low. *ZjEXPA6* and *ZjEXPA12*

Table 2 Number of variations of *cis*-acting elements in the promoter region of each jujube expansin gene

Gene name	Development-related elements	Environmental stress-related elements	Hormone-responsive elements	Light-responsive elements	Promoter-related elements	Site-binding related elements	Others
<i>ZjEXPA1</i>	3	5	5	14	2	0	8
<i>ZjEXPA2</i>	5	3	3	9	2	0	2
<i>ZjEXPA3</i>	5	4	3	5	4	0	6
<i>ZjEXPA4</i>	3	2	3	6	4	1	5
<i>ZjEXPA5</i>	5	4	2	11	4	0	6
<i>ZjEXPA6</i>	2	5	5	11	3	1	12
<i>ZjEXPA7</i>	2	5	4	8	2	1	4
<i>ZjEXPA8</i>	3	3	4	17	2	0	5
<i>ZjEXPA9</i>	2	3	5	11	4	1	7
<i>ZjEXPA10</i>	2	3	1	10	4	1	4
<i>ZjEXPA11</i>	7	3	4	10	4	0	8
<i>ZjEXPA12</i>	3	3	1	11	4	1	10
<i>ZjEXPA13</i>	1	4	1	5	4	2	3
<i>ZjEXPA14</i>	2	4	5	11	3	0	8
<i>ZjEXPA15</i>	2	5	4	12	3	1	5
<i>ZjEXPA16</i>	2	4	4	12	3	1	4
<i>ZjEXPA17</i>	2	6	3	11	3	1	2
<i>ZjEXPA18</i>	4	4	5	9	4	0	3
<i>ZjEXPA19</i>	3	5	0	10	4	1	6
<i>ZjEXPB1</i>	1	4	5	8	3	0	7
<i>ZjEXPB2</i>	4	4	1	11	5	1	4
<i>ZjEXPB3</i>	1	2	5	9	4	1	7
<i>ZjEXLA1</i>	3	5	1	11	2	1	3
<i>ZjEXLB1</i>	3	4	4	8	5	1	5
<i>ZjEXLB2</i>	2	2	3	9	6	0	4
<i>ZjEXLB3</i>	2	5	4	15	4	1	6
<i>ZjEXLB4</i>	3	3	3	12	3	0	10
<i>ZjEXLB5</i>	2	4	1	9	3	2	5
<i>ZjEXLB6</i>	3	3	2	9	2	1	6
<i>ZjEXLB7</i>	1	4	1	10	3	1	3

All *cis*-acting elements were divided into seven categories according to their function

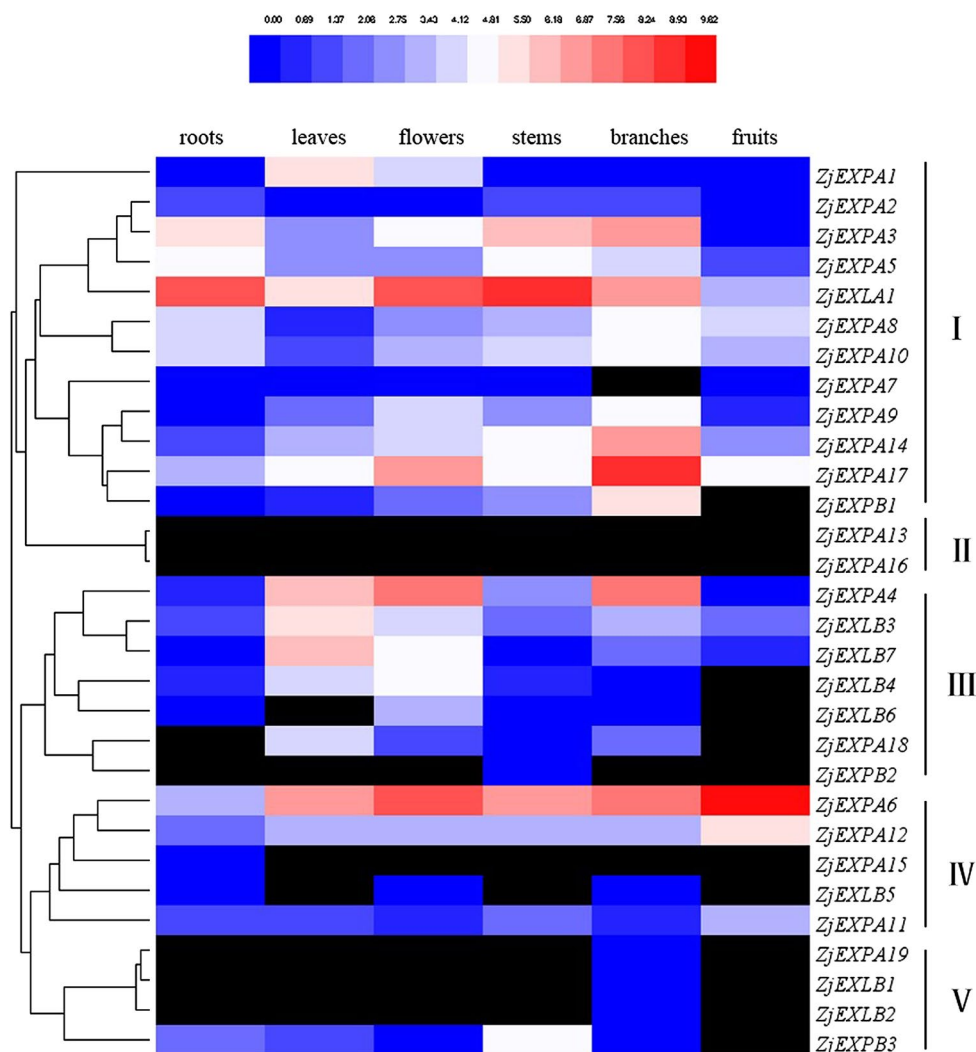
displayed higher expression levels in fruits, thus indicating their important roles in fruit development. *ZjEXLB1*, *ZjEXLB2*, and *ZjEXPA19*, which were clustered together, were only expressed in branches.

Expression profiles of expansin genes in different jujube tissues analyzed by qRT-PCR

To comprehensively understand the physiological functions of jujube expansin genes, we analyzed the expressions of 14 randomly selected expansin genes in jujube under normal growth conditions in eight different tissues by qRT-PCR. Expression levels of the 14 *ZjEXP* genes differed in these tissues, indicating their involvement in the development of one or more tissues (Fig. 8). *ZjEXPA7* was dominantly expressed in young leaves, but was minimally expressed in

other seven tissues. The expression of *ZjEXPA2*, *ZjEXPA8*, *ZjEXPA9*, and *ZjEXPA10* was notably high in shoots, whereas *ZjEXPA3*, *ZjEXPA5*, and *ZjEXPA17* showed high expressions in flowers. In terms of different jujube fruit developmental stages, analyzed expansin family members, except for *ZjEXPA7* and *ZjEXPA9*, were all expressed at high levels during at least one stage of fruit development. *ZjEXPA11* and *ZjEXPA12* were relatively highly expressed in full-red fruits, whereas the other ten genes were primarily expressed in young fruits, with their expressions gradually decreasing with fruit maturity. Among them, *ZjEXPA4* and *ZjEXPA6* had an expression peak in young fruits and were up-regulated in half-red fruits, suggesting their importance in fruit ripening. These results further highlighted that expansin genes were involved in jujube vegetative and reproductive growth, especially in fruit development.

Fig. 7 Expression profile of 30 jujube expansin genes in different tissues, subdivided into groups (labelled I–V) based on the transcriptome data. The legend represents the logarithmic normalized FPKM. Black means that the raw FPKM value was zero



Discussion

Systematic analyses have previously been conducted to identify and characterize expansin families in several models, crop and fruit plant species. In this study, 30 jujube expansin genes containing two conserved domains were first identified in jujube genome through genome-wide analysis. Analysis and comparison of the sizes of expansin subfamilies in five species (*Arabidopsis*, rice, apple, grape, and jujube) revealed an uneven distribution of each gene subfamily among species. For example, a monocot rice contains more EXPB members than does eudicots (i.e., *Arabidopsis*, apple, grape, and jujube)—an observation consistent with the results of the previous studies (Zhang et al. 2014a; Li et al. 2016) and possibly due to differences in cell-wall components and evolution among species (Sarkar et al. 2009; Dal Santo et al. 2013; Zhu et al. 2014). The EXLB subfamily has apparently expanded in woody species, with four members in grape, four in apple and seven in jujube, but not in the herbaceous plants rice and *Arabidopsis*, which each have only one EXLB

gene; this indicates that these genes may have a unique function in woody plants, such as increasing the adaptability to certain environments (Zhu et al. 2014; Ding et al. 2016). Phylogenetic analysis of expansin subfamilies revealed that the majority of jujube expansin genes appear more closely related to grape and apple expansin genes, while those of rice and *Arabidopsis* are more closely related, thus implying that there was a set of ancestral expansin genes corresponding to each subfamily predating the divergence of monocot and eudicot (Valdivia et al. 2007). Within the same phylogenetic subfamily and even subgroup, members mostly have a conserved gene structure, thus confirming their close evolutionary relationship and classification (Zhang et al. 2014a).

To understand the regulation of *ZjEXP* expression, we used the PlantCARE program to analyze *cis*-acting elements in all *ZjEXP* promoter regions. Four classes of *cis*-acting elements, namely, development-related elements, environmental stress-related elements, hormone-responsive elements, and light-responsive elements, appear to be significantly abundant in *ZjEXPs*. In plants, both

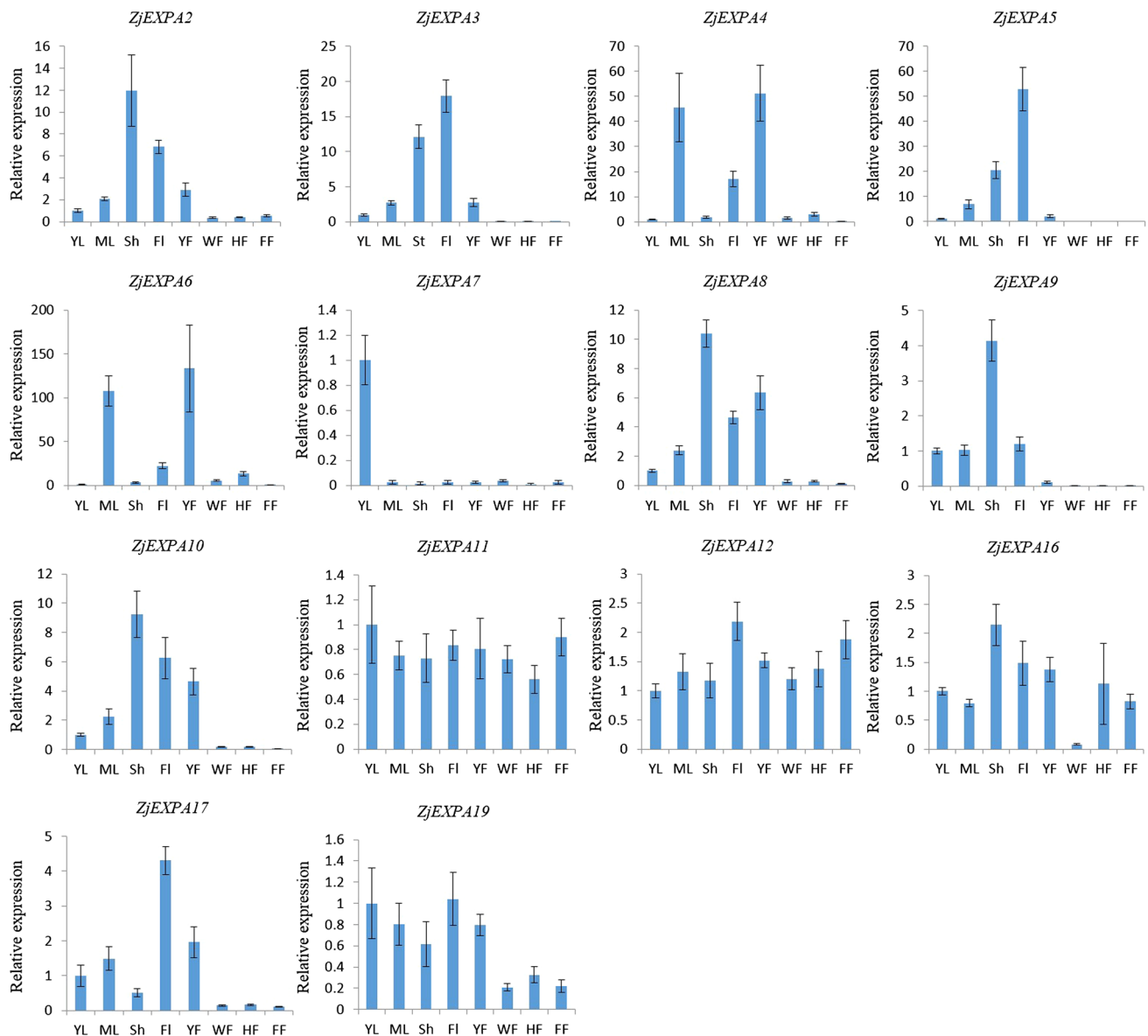


Fig. 8 qRT-PCR analysis for 14 jujube expansin genes in eight different tissues. YL young leaves, ML mature leaves, Sh shoots, FI flowers, YF young fruits, WF white-mature fruits, HF half-red fruits, FF

full-red fruits. The mean expression value was calculated from three independent replicates. The vertical bars show the standard deviation

environmental and internal factors can affect the expression of expansin genes, which leads them to involve in various developmental processes conjectured to be regulated by the corresponding *cis*-acting elements contained within them. Two leaf development-related *cis*-acting elements (HD-Zip1 and HD-Zip2) are present in the *ZjEXPA11* promoter (Table S3), which is consistent with its relatively high expression in young leaves (Fig. 8). We also found an ethylene response element (ERE) in *ZjEXPA6* promoter regions that was highly expressed in mature leaves and young fruits, which suggests that *ZjEXPA6* plays key roles in plant maturity and senility. Plant hormones and

environmental stimuli that can regulate the expression of expansin genes have been reported in several plants. For example, *SlEXPA3*, *SlEXPA7*, and *SlEXPA9* in tomato were induced by the application of exogenous ABA, IAA, and heat stress (Lu et al. 2016), while the poplar expansin gene *PtEXPA8* was significantly up-regulated in leaves under heat stress (Liu et al. 2016). Different types and numbers of regulatory elements are present in distinct *ZjEXP* promoters, thus indicating that *ZjEXP* genes may be regulated by different environmental and internal factors and have various functional roles in different developmental processes related to cell-wall modification.

According to the results of our transcriptome and qRT-PCR analyses, *ZjEXPs* are expressed in various tissues during different developmental periods, which indicate their possible direct or indirect involvement in certain developmental stages. As inferred from both the qRT-PCR and transcriptome data, *ZjEXPA11* exhibited constitutive expression in all tissues. In terms of the transcriptome analysis, *ZjEXPA8*, *ZjEXPA9*, and *ZjEXPA10* were highly expressed in branches, whereas these three genes were mainly expressed in shoots in the qRT-PCR analysis. We conjecture that branches in the transcriptome were essentially equivalent to shoots in our study. Although some homologous gene pairs in the same subgroup, such as *ZjEXPA4* and *ZjEXPA6*, had similar expression profiles in qRT-PCR analysis, distinction was found in transcriptome analysis. These inconsistencies between transcriptome and qRT-PCR analysis may be due to lack of exact information for development stages of tissues employed in the transcriptome data (Song et al. 2017).

The involvement of expansins in plant development as well as defense and abiotic stress responses has been proven in several plants (Che et al. 2016; Kuluev et al. 2017; Perini et al. 2017). Analysis of phylogenetic and homologous relationships and gene expression profiles can provide important information for gene function. Based on our phylogenetic tree, *ZjEXPA6* is closely related to *MdEXP4* in the EXPA-III subgroup, the latter playing a major role during apple fruit ontogeny (Wakasa et al. 2003). Our qRT-PCR analysis indicated that *ZjEXPA6* has higher expression levels in mature leaves, flowers, and half-red fruits, especially young fruits, which suggests the potentially important role of this gene in fruit development and maturity. In our phylogenetic tree, *ZjEXPA14* and *AtEXPA10* are placed in the EXPA-I subgroup. *AtEXPA10* influences the sizes of vegetative organs (Kuluev et al. 2012) and exerts its regulatory roles in leaf growth and pedicel abscission (Cho and Cosgrove 2000). *ZjEXPA9* is closely related to *AtEXPA5*, with both belonging to the EXPA-I subgroup. *AtEXPA5* regulates plant growth, including the elongation of roots, seedling hypocotyls, and rosette leaves (Park et al. 2011), and *ZjEXPA14* and *ZjEXPA9* were expressed in leaves, branches, and flowers in our study. In addition, *AtEXPA7* and *AtEXPA18* genes are required for root hair tip growth (Cho and Cosgrove 2002; Lin et al. 2011). According to the phylogenetic tree, *ZjEXPA15*, *ZjEXPA16*, *AtEXPA7*, and *AtEXPA18* belong to the EXPA-X subgroup. Our homology analysis revealed that *ZjEXPA15* is homologous to *AtEXPA7*, while *ZjEXPA16* is homologous to *AtEXPA18*. Significantly, *ZjEXPA15* is only expressed in roots according to the transcriptome analysis. These results indicate that the above four genes may play a similar role in root development. The only member of the EXLA-I subgroup in jujube is *ZjEXLA1*, whose homolog in *Arabidopsis*, *AtEXLA2*, appears to be important in the pathogenesis of necrotrophic pathogens and in tolerance to abiotic

stress (Abuqamar et al. 2013). *ZjEXPA17* and *AtEXPA3* are a homologous gene pair in the EXPA-IV subgroup, while homologs *ZjEXPB1* and *AtEXPB1* belong to the EXPB-II subgroup. *AtEXPA3* and *AtEXPB1* have been proven to enhance salt stress and hormone sensitivity in *Arabidopsis* (Kwon et al. 2008).

In this study, we performed a complete genome-wide analysis of the expansin family in jujube and generated a wide range of expression data. Our results provide valuable information for the further elucidation of the evolution and divergence of plant expansin genes. The study findings may also aid understanding of the biological functions of jujube expansin genes and the molecular basis of many agriculturally important jujube traits.

Author contribution statement YL conceived and designed the experiments. LH, ZZ, SD, and YZ conducted the experiments, and LH and ZZ collected and analyzed the data. LH wrote and revised the manuscript. MP and YL provided valuable suggestions on the manuscript. YL obtained funding and is responsible for this article. All authors read and approved the final manuscript.

Acknowledgements This work was supported by grants from the Fundamental Research Funds for the Central Universities (2016ZCQ05) and Major Science and Technology Special Project of Xuchang, Henan province (20170112006).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

- Abuqamar S, Ajeb S, Sham A, Enan MR, Iratni R (2013) A mutation in the expansin-like A2 gene enhances resistance to necrotrophic fungi and hypersensitivity to abiotic stress in *Arabidopsis thaliana*. Mol Plant Pathol 14(8):813–827
- Artimo P, Jonnalagedda M, Arnold K, Baratin D, Csardi G, de Castro E, Duvaud S, Flegel V, Fortier A, Gasteiger E, Grosdidier A, Hernandez C, Ioannidis V, Kuznetsov D, Liechti R, Moretti S, Mostaguir K, Redaschi N, Rossier G, Xenarios I, Stockinger H (2012) ExPASy: SIB bioinformatics resource portal. Nucleic Acids Res 40:W597–W603
- Bailey TL, Boden M, Buske FA, Frith M, Grant CE, Clementi L, Ren J, Li WW, Noble WS (2009) MEME SUITE: tools for motif discovery and searching. Nucleic Acids Res 37:W202–W208
- Boron AK, Van Loock B, Suslov D, Markakis MN, Verbelen JP, Visserberg K (2015) Over-expression of *AtEXLA2* alters etiolated *Arabidopsis* hypocotyl growth. Ann Bot 115(1):67–80
- Bu J, Zhao J, Liu M (2016) Expression stabilities of candidate reference genes for RT-qPCR in Chinese jujube (*Ziziphus jujuba* Mill.) under a variety of conditions. PLoS One 11(4):e0154212

- Che J, Yamaji N, Shen RF, Ma JF (2016) An Al-inducible expansin gene, *OsEXPA10* is involved in root cell elongation of rice. *Plant J* 88(1):132–142
- Chen Y, Han Y, Zhang M, Zhou S, Kong X, Wang W (2016) Overexpression of the wheat expansin gene *TaEXPA2* improved seed production and drought tolerance in transgenic tobacco plants. *PLoS One* 11(4):e0153494
- Cho HT, Cosgrove DJ (2000) Altered expression of expansin modulates leaf growth and pedicel abscission in *Arabidopsis thaliana*. *Proc Natl Acad Sci USA* 97(17):9783–9788
- Cho HT, Cosgrove DJ (2002) Regulation of root hair initiation and expansin gene expression in *Arabidopsis*. *Plant Cell* 14(2):3237–3253
- Cosgrove DJ (2000) Loosening of plant cell walls by expansins. *Nature* 407(6802):321–326
- Cosgrove DJ (2005) Growth of the plant cell wall. *Nat Rev Mol Cell Biol* 6(11):850–861
- Cosgrove DJ (2015) Plant expansins: diversity and interactions with plant cell walls. *Curr Opin Plant Biol* 25:162–172
- Cosgrove DJ, Bedinger P, Durachko DM (1997) Group I allergens of grass pollen as cell wall-loosening agents. *Proc Natl Acad Sci USA* 94(10):6559–6564
- Dal Santo S, Vannozzi A, Toriell GB, Fasoli M, Venturini L, Pezzotti M, Zenoni S (2013) Genome-wide analysis of the expansin gene superfamily reveals grapevine-specific structural and functional characteristics. *PLoS One* 8(4):e62206
- Ding A, Marowa P, Kong Y (2016) Genome-wide identification of the expansin gene family in tobacco (*Nicotiana tabacum*). *Mol Genet Genomics* 291(5):1891–1907
- Finn RD, Attwood TK, Babbitt PC, Bateman A, Bork P, Bridge AJ, Chang HY, Dosztányi Z, El-Gebali S, Fraser M, Gough J, Haft D, Holliday GL, Huang H, Huang X, Letunic I, Lopez R, Lu S, Marchler-Bauer A, Mi H, Mistry J, Natale DA, Necci M, Nuka G, Orengo CA, Park Y, Pesseat S, Piovesan D, Potter SC, Rawlings ND, Redaschi N, Richardson L, Rivoire C, Sangrador-Vegas A, Sigrist C, Sillitoe I, Smithers B, Squizzato S, Sutton G, Thanki N, Thomas PD, Tosatto SC, Wu CH, Xenarios I, Yeh LS, Young SY, Mitchell AL (2017) InterPro in 2017—beyond protein family and domain annotations. *Nucleic Acids Res* 45(D1):D190–D199
- Gao QH, Wu CS, Wang M (2013) The jujube (*Ziziphus jujuba* Mill.) fruit: a review of current knowledge of fruit composition and health benefits. *J Agric Food Chem* 61(14):3351–3363
- Georgelis N, Nikolaidis N, Cosgrove DJ (2015) Bacterial expansins and related proteins from the world of microbes. *Appl Microbiol Biotechnol* 99(9):3807–3823
- Guimaraes LA, Mota APZ, Araujo ACG, de Alencar Figueiredo LF, Pereira BM, de Passos Saraiva MA, Silva RB, Danchin EGJ, Guimaraes PM, Brasileiro ACM (2017) Genome-wide analysis of expansin superfamily in wild *Arachis* discloses a stress-responsive expansin-like B gene. *Plant Mol Biol* 94(1–2):79–96
- Hu B, Jin JP, Guo AY, Zhang H, Luo JH, Gao G (2015) GSDS 2.0: an upgraded gene feature visualization server. *Bioinformatics* 31(8):1296–1297
- Huang J, Zhang C, Zhao X, Fei Z, Wan K, Zhang Z, Pang X, Yin X, Bai Y, Sun X, Gao L, Li R, Zhang J, Li X (2016) The jujube genome provides insights into genome evolution and the domestication of sweetness/acidity taste in fruit trees. *PLoS Genet* 12(12):e1006433
- Kende H, Bradford K, Brummell D, Cho HT, Cosgrove D, Fleming A, Gehring C, Lee Y, McQueen-Mason S, Rose J, Voesenek LA (2004) Nomenclature for members of the expansin superfamily of genes and proteins. *Plant Mol Biol* 55(3):311–314
- Krzywinski M, Schein J, Birol I, Connors J, Gascoyne R, Horsman D, Jones SJ, Marra MA (2009) Circos: an information aesthetic for comparative genomics. *Genome Res* 19(9):1639–1645
- Kuluev BR, Knyazev AB, Lebedev YP, Chemeris AV (2012) Morphological and physiological characteristics of transgenic tobacco plants expressing expansin genes: *AtEXP10* from *Arabidopsis* and *PnEXPA1* from poplar. *Russ J Plant Physiol* 59(1):97–104
- Kuluev BR, Knyazev AV, Mikhaylova EV, Chemeris AV (2017) The role of expansin genes *PtrEXPA3*, and *PnEXPA3*, in the regulation of leaf growth in poplar. *Russ J Plant Physiol* 53(6):651–660
- Kwon YR, Lee HJ, Kim KH, Hong SW, Lee SJ, Lee H (2008) Ectopic expression of *Expansin3* or *Expansinβ1* causes enhanced hormone and salt stress sensitivity in *Arabidopsis*. *Biotechnol Lett* 30(7):1281–1288
- Lee Y, Choi DS, Kende H (2001) Expansins: ever-expanding numbers and functions. *Curr Opin Plant Biol* 4(6):527–532
- Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombauts S (2002) PlantCARE, a database of plant *cis*-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. *Nucleic Acids Res* 30(1):325–327
- Li B, Dewey CN (2011) RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC Bioinformatics* 12:323
- Li L, Stoeckert CJ, Roos DS (2003) OrthoMCL: identification of ortholog groups for eukaryotic genomes. *Genome Res* 13(9):2178–2189
- Li N, Pu Y, Gong Y, Yu Y, Ding H (2016) Genomic location and expression analysis of expansin gene family reveals the evolutionary and functional significance in *Triticum aestivum*. *Genes Genom* 38(11):1021–1030
- Lin C, Choi HS, Cho HT (2011) Root hair-specific EXPANSIN A7 is required for root hair elongation in *Arabidopsis*. *Mol Cells* 31(4):393–397
- Liu MJ (2003) Genetic diversity of Chinese jujube (*Ziziphus jujuba* Mill.). *Acta Hort* 623(623):351–355
- Liu MJ (2006) Chinese jujube: botany and horticulture. *Hortic Rev* 32:229–298
- Liu M, Wang M (2009) Chinese jujube germplasm resources. Forestry Publishing House, Beijing
- Liu M, Zhao J, Cai Q, Liu G, Wang J, Zhao Z, Liu P, Dai L, Yan G, Wang W, Li X, Chen Y, Sun Y, Liu Z, Lin M, Xiao J, Chen Y, Li X, Wu B, Ma Y, Jian J, Yang W, Yuan Z, Sun X, Wei Y, Yu L, Zhang C, Liao S, He R, Guang X, Wang Z, Zhang Y, Luo L (2014) The complex jujube genome provides insights into fruit tree biology. *Nat Commun* 5:5315
- Liu H, Li H, Zhang H, Li J, Xie B, Xu J (2016) The expansin gene *PttEXPA8* from poplar (*Populus tomentosa*) confers heat resistance in transgenic tobacco. *Plant Tiss Org Cult* 126(2):353–359
- Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using realtime quantitative PCR and the $2^{-\Delta\Delta C_T}$ method. *Methods* 25(4):402–408
- Lu ZM, Liu K, Yan ZX, Li XG (2010) Research status of nutrient component and health functions of *Ziziphus jujuba* Mill. *Acta Hort* Sin 37(12):2017–2024
- Lu YG, Liu LF, Wang X, Han ZH, Ouyang B, Zhang JH, Li HX (2016) Genome-wide identification and expression analysis of the expansin gene family in tomato. *Mol Genet Genomics* 291(2):597–608
- Palapol Y, Kunyamee S, Thongkhum M, Ketsa S, Ferguson IB, van Doorn WG (2015) Expression of expansin genes in the pulp and the dehiscence zone of ripening durian (*Durio zibethinus*) fruit. *J Plant Physiol* 182:33–39
- Park CH, Kim TW, Son SH, Hwang JY, Lee SC, Chang SC, Kim SH, Kim SW, Kim SK (2011) Brassinosteroids control *AtEXPA5* gene expression in *Arabidopsis thaliana*. *Phytochemistry* 71(4):380–387
- Perini MA, Sin IN, Villarreal NM, Marina M, Powell AL, Martínez GA, Civello PM (2017) Overexpression of the carbohydrate binding module from *Solanum lycopersicum* expansin 1 (*Sl-EXPI*)

- modifies tomato fruit firmness and *Botrytis cinerea* susceptibility. *Plant Physiol Biochem* 113:122–132
- Saito T, Tuan PA, Katsumi-Horigane A, Bai S, Ito A, Sekiyama Y, Ono H, Moriguchi T (2015) Development of flower buds in the Japanese pear (*Pyrus pyrifolia*) from late autumn to early spring. *Tree Physiol* 35(6):653–662
- Sampedro J, Cosgrove DJ (2005) The expansin superfamily. *Genome Biol* 6(12):242. <https://doi.org/10.1186/gb-2005-6-12-242>
- Sarkar P, Bosneaga E, Auer M (2009) Plant cell walls throughout evolution: towards a molecular understanding of their design principles. *J Exp Bot* 60(13):3615–3635
- Song S, Zhou H, Sheng S, Cao M, Li Y, Pang X (2017) Genome-wide organization and expression profiling of the SBP-box gene family in Chinese jujube (*Ziziphus jujuba* Mill.). *Int J Mol Sci* 18(8):1734
- Tamura K, Stecher G, Peterson D, Filipski A, Kumar S (2013) MEGA6: molecular evolutionary genetics analysis version 6.0. *Mol Biol Evol* 30(12):2725–2729
- Thompson JD, Higgins DG, Gibson TJ (1994) CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Res* 22(22):4673–4680
- Tovar-Herrera OE, Batista-Garcia RA, Sanchez-Carbente Mdel R, Iracheta-Cardenas MM, Arevalo-Nino K, Folch-Mallol JL (2015) A novel expansin protein from the white-rot fungus *Schizophyllum commune*. *PLoS One* 10(3):e0122296
- Trapnell C, Roberts A, Goff L, Pertea G, Kim D, Kelley DR, Pimentel H, Salzberg SL, Rinn JL, Pachter L (2012) Differential gene and transcript expression analysis of RNA-Seq experiments with HOPHAT and cufflinks. *Nat Protoc* 7(3):562–578
- Tsuchiya M, Satoh S, Iwai H (2015) Distribution of XTH, expansin, and secondary-wall-related CesA in floral and fruit abscission zones during fruit development in tomato (*Solanum lycopersicum*). *Front Plant Sci* 6:323
- Valdivia ER, Sampedro J, Lamb JC, Chopra S, Cosgrove DJ (2007) Recent proliferation and translocation of pollen group 1 allergen genes in the maize genome. *Plant Physiol* 143(3):1269–1281
- Wakasa Y, Hatsuyama Y, Takahashi A, Sato T, Niizeki M, Harada T (2003) Divergent expression of six expansin genes during apple fruit ontogeny. *Eur J Hort Sci* 68(6):253–259
- Yan A, Wu M, Yan L, Hu R, Ali I, Gan Y (2014) *AtEXP2* is involved in seed germination and abiotic stress response in *Arabidopsis*. *PLoS One* 9(1):e85208
- Yennawar NH, Li LC, Dudzinski DM, Tabuchi A, Cosgrove DJ (2006) Crystal structure and activities of EXPB1 (*Zea m 1*), a β -expansin and group-1 pollen allergen from maize. *Proc Natl Acad Sci USA* 103(40):14664–14671
- Zhang S, Xu R, Gao Z, Chen C, Jiang Z, Shu H (2014a) A genome-wide analysis of the expansin genes in *Malus $\times$ Domestica*. *Mol Genet Genom* 289(2):225–236
- Zhang W, Yan H, Chen W, Liu J, Jiang C, Jiang H, Zhu S (2014b) Genome-wide identification and characterization of maize expansin genes expressed in endosperm. *Mol Genet Genomics* 289(6):1061–1074
- Zhu Y, Wu N, Song W, Yin G, Qin Y, Yan Y, Hu Y (2014) Soybean (*Glycine max*) expansin gene superfamily origins: segmental and tandem duplication events followed by divergent selection among subfamilies. *BMC Plant Biol* 14:93